Table 5.1 Fat soluble vitamins		
Vitamins	**Functions**	**Symptoms of Deficiency**
A (Retinol)/ Antixerophthalmic vitamin	Plays a vital role in visual perception. Maintenance and growth of epithelial tissue.	Night blindness **(Nyctalopia)**, Xerophthalmia (drying of eyeballs), **Bitot's spot** in the cornea, **Dermatosis** (dry and scaly skin) and **Keratomalacia** Atrophy of lacrymal glands and reduction in tear secretion
D (Calciferol)/ Antirachitic vitamin	Promotes intestinal absorption of calcium and phosphorus. Formation of teeth and bones.	**Rickets** in children (softness and deformities of bones and bow legs and pigeon chest) and **Osteomalacia** in adults (weak and fragile bones, bent, deformed pelvis).
E (Tocopherol) / Antisterility vitamin	Antioxidant It keeps the skin healthy by reducing the process of ageing.	**Sterility** in animals, Ruptured red blood cells
K Anti haemorrahagic vitamin.	Required for the synthesis of prothrombin in the liver.	Defect in blood clotting called Haemorrhagic manifestations.

Table 5.2 Water soluble vitamins		
Vitamins	**Functions**	**Symptoms of Deficiency**
B_1 (Thiamine)	Involved in carbohydrate metabolism. Act as a coenzyme	**Beriberi:** affects muscular, nervous and cardiovascular system
B_2 (Riboflavin)	Acts as coenzyme in oxidation and reduction reactions	Inflammation, soreness and fissures in the corners of the mouth, lips and tongue. Loss of appetite. Skin and eye disorder.
B_3 (Pantothenic acid)	Acts as coenzyme A and is essential for the metabolism of fats and carbohydrates	Gastrointestinal disorders, anaemia, Burning feet syndrome, etc.
B_4 (Choline)	Precursor for acetylcholine	Fatty liver.
B_5 (Niacin / Nicotinic acid)	Derivatives of coenzymes	**Pellagra** (4D Syndrome) characterised by dermatitis, diarrhoea and dementia (mental deterioration) and death.
B_6 (Pyridoxine)	Haemoglobin formation, brain, heart and liver activities	**Dermatitis,** convulsions, muscular twitching and **anaemia**

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